

A PROJECTIVE APPROACH TO NON-DIVIDING SETS

THEOFIL XEFF

ABSTRACT. Let $F(N)$ be the largest size of a set $A \subseteq \{1, \dots, N\}$ in which no element divides the sum of any nonempty subset of the remaining elements. Erdős, Lev, Rauzy, Sándor and Sárközy proved $F(N) = O(N^{1/2})$, and constructions of Erdős give $F(N) \gg N^{1/5}$. We prove the matching upper bound:

$$F(N) = N^{1/5+o(1)}.$$

The key new idea is a projective normalization $v \mapsto v/\lambda(v)$. Divisibility constraints depend only on the values of the linear functional λ , so they survive the normalization, while the associated convex geometry drops by one dimension; the density-increment method behind Pham and Zakharov's sharp non-averaging bound then applies one dimension lower, moving the exponent from $1/4$ to $1/5$. The proof has been formally verified in Lean 4.

1. INTRODUCTION

For a finite set $A \subseteq \mathbb{Z}_{>0}$, write

$$\Sigma(A) = \left\{ \sum_{a \in S} a : S \subseteq A \right\}$$

for its set of subset sums.

Definition 1.1. A finite set $A \subseteq \mathbb{Z}_{>0}$ is *non-dividing* if, for every $a \in A$ and every nonempty set $S \subseteq A \setminus \{a\}$,

$$a \nmid \sum_{s \in S} s.$$

Let $F(N)$ be the maximum cardinality of a non-dividing subset of $[N] = \{1, \dots, N\}$.

Erdős, Lev, Rauzy, Sándor and Sárközy introduced this problem and proved the bound $F(N) < 3N^{1/2} + 1$ [4]; it appears as Problem 131 in the collection of Erdős problems maintained by Bloom [1]. The previously known polynomial estimates were

$$N^{1/5} \ll F(N) \leq N^{1/4+o(1)}.$$

The lower bound is implicit in the constructions of Erdős and of Erdős–Lev–Rauzy–Sándor–Sárközy [3, 4]. The upper bound follows because every non-dividing set is non-averaging, together with the sharp theorem of Pham and Zakharov [5].

Our main result closes the gap between these exponents.

Theorem 1.2. *One has*

$$F(N) = N^{1/5+o(1)}.$$

1.1. Outline of the proof. Every non-dividing set is non-averaging: an averaging relation $\sum_{s \in S} s = |S|a$ is already forbidden modulo a . The previous upper bound $F(N) \leq N^{1/4+o(1)}$ came from this observation alone, combined with the sharp non-averaging theorem of Pham–Zakharov [5]. Since the exponent $1/4$ is best possible for non-averaging sets, any further progress must use the forbidden divisibilities in an essential way, and the difficulty is that divisibility, unlike averaging, is destroyed by translation.

Our starting point is that divisibility survives a different family of operations. A dyadic pigeonhole places a large part of the set in a shell $X \leq a \leq 2X$, at the cost of a logarithmic factor. We then work in $\mathbb{Z}^d$, with a set V in a symmetric box B and an integer linear functional λ taking values in $[X, 2X]$ on V , and forbid $\lambda(a) \mid \sum_{v \in S} \lambda(v)$ for $a \in V$ and nonempty $S \subseteq V \setminus \{a\}$. Because these relations see only the values of λ , they are preserved exactly when we pass to a subset or apply the coefficient identification of a proper homogeneous generalized arithmetic progression (GAP) containing V ; these two operations turn out to be all we need.

The structural input is the homogeneous subset-sum theorem of Conlon–Fox–Pham [2], in the form recorded in [5], which places V in a proper GAP whose subset sums fill a large translated copy of that GAP. Iterating coefficient identifications (Section 4), we may assume the pair (V, B) is *irreducible*: every large subset of V has full structural dimension, and its structural GAP occupies a fixed proportion of the box.

The new idea enters here. Normalizing $\pi(v) = v/\lambda(v)$ sends the shell into the affine hyperplane $\{\lambda = 1\}$, of dimension $d-1$, and we prove (Theorem 6.4) that for an irreducible pair the projected family $\pi(V)$ must be in approximate convex position. Suppose it is not. A duality argument produces some $a \in V$ and a bounded positive representation $Ta = \sum_{v \neq a} \theta_v v$ by the remaining elements; zonotope rounding converts this identity into genuine subset sums, which fill a *tube*: every point of a small-index structural lattice in a box around Ta . The index is small enough that some positive multiple $m_0 a$ lands in the tube, making $m_0 \lambda(a)$ a subset sum of the values $\lambda(v)$ with $v \neq a$ (exactly a forbidden relation).

The rest is a density increment in the style of Pham–Zakharov, run one dimension lower. Their convex-density lemma, applied in the hyperplane, finds a small cap containing many of the points $\pi(v)$, and lifting the cap back to the lattice (Lemma 7.2) shrinks the box by the volume of the cap. This gain of one dimension is what moves the exponent: setting $\sigma_1 = \frac{1}{5}$, $\sigma_2 = \frac{1}{4}$ and $\sigma_d = \frac{d-2}{d}$ for $d \geq 3$, so that for $d \geq 2$ the number σ_d is the sharp non-averaging exponent in dimension $d-1$, the iteration (Theorem 8.2) shows that a shell non-dividing set satisfies $|V| \leq |B|^{\sigma_d + o(1)}$. The case $d = 1$ is Theorem 1.2.

1.2. Formal verification. The complete proof of Theorem 1.2 has been formalized and machine-checked in Lean 4 over the mathlib library [7]. The formalization follows the Comparator workflow of the Lean Focused Research Organization: a short trusted file states the two definitions and the main theorem, and the kernel certifies that the development proves exactly that statement. The only non-foundational axioms are the statements of the quoted literature inputs — the structure theorem of Conlon–Fox–Pham in the form of [5, Corollary 5], the discrete John lemma [5, Lemma 7], zonotope rounding [5, Lemma 13], the convex-density lemma [5, Lemma 1], and the Blaschke selection and volume continuity theorems from [6] — so the formalization verifies precisely the reduction carried out in this paper. In view of this, we abbreviate some routine uniformity and bookkeeping verifications in the longer proofs below; every abbreviated step is carried out in complete detail in the formalization.

2. ELEMENTARY STRUCTURE AND THE LOWER BOUND

The bound $F(N) \gg N^{1/5}$ is essentially known, being implicit in the constructions of [3, 4]; we include a short self-contained proof.

2.1. A translation lemma. A finite set $B \subseteq \mathbb{Z}$ is *non-averaging* if there are no $b \in B$ and nonempty $S \subseteq B \setminus \{b\}$ such that

$$\sum_{s \in S} s = |S|b.$$

Lemma 2.1. *Let $B \subseteq \mathbb{Z}$ be a non-averaging set of cardinality m and diameter D . If*

$$x + \min B > mD,$$

then $x + B$ is non-dividing.

Proof. Suppose that, for $b_j \notin S \subseteq B$,

$$x + b_j \mid \sum_{b_i \in S} (x + b_i).$$

Put $s = |S|$. Subtracting $s(x + b_j)$, we see that $x + b_j$ divides

$$R = \sum_{b_i \in S} (b_i - b_j).$$

Since $|R| \leq sD < mD < x + \min B \leq x + b_j$, one has $R = 0$. This is an averaging relation in B , a contradiction. $\square$

2.2. An explicit $N^{1/5}$ construction.

Proposition 2.2. *For every integer $m \geq 2$, there is a non-dividing set of cardinality m contained in $[2m^5]$. Consequently,*

$$F(N) \gg N^{1/5}.$$

Proof. Put

$$C = m^3 + 1, \quad b_i = Ci + i^2 \quad (1 \leq i \leq m),$$

and let $B = \{b_1, \dots, b_m\}$. Suppose that

$$\sum_{i \in S} b_i = |S|b_j, \quad j \notin S.$$

Writing $s = |S|$ gives

$$C \left(\sum_{i \in S} i - sj \right) + \left(\sum_{i \in S} i^2 - sj^2 \right) = 0.$$

The second parenthesis is an integer of absolute value at most $m^3 < C$. Reducing the displayed identity modulo C therefore shows that this parenthesis is zero; the first parenthesis is then zero as well. Hence

$$\sum_{i \in S} (i - j)^2 = 0,$$

which is impossible because $j \notin S$. Thus B is non-averaging.

Let $\Delta = b_m - b_1$ and set

$$x = m\Delta - b_1 + 1, \quad A = x + B.$$

Then $\min A = m\Delta + 1$, so Lemma 2.1 implies that A is non-dividing. Moreover,

$$\Delta = (m^3 + 1)(m - 1) + m^2 - 1 = m^4 - m^3 + m^2 + m - 2,$$

and

$$\max A = (m + 1)\Delta + 1 < 2m^5$$

for $m \geq 2$. $\square$

3. THE PROJECTIVE FRAMEWORK

The upper-bound proof is most naturally formulated in boxes in $\mathbb{Z}^d$.

Definition 3.1 (Shell non-dividing set). Let $B \subseteq \mathbb{Z}^d$ be a symmetric axis-parallel box, let $\lambda : \mathbb{Z}^d \rightarrow \mathbb{Z}$ be a nonzero integer linear functional, and let $X > 0$. A finite set $V \subseteq B$ is (λ, X) -shell non-dividing if

$$X \leq \lambda(v) \leq 2X \quad (v \in V)$$

and, for every $a \in V$ and every nonempty $S \subseteq V \setminus \{a\}$,

$$\lambda(a) \nmid \sum_{v \in S} \lambda(v).$$

The shell restriction loses only a logarithmic factor in the original one-dimensional problem.

Definition 3.2 (Projective normalization). For v in the positive λ -shell, define

$$\pi(v) = \frac{v}{\lambda(v)}.$$

Then $\pi(v)$ belongs to the affine hyperplane

$$H_\lambda = \{y \in \mathbb{R}^d : \lambda(y) = 1\},$$

which has dimension $d - 1$.

This is the source of the expected dimension drop.

3.1. The exponent sequence. Define

$$\sigma_1 = \frac{1}{5}, \quad \sigma_2 = \frac{1}{4}, \quad \sigma_d = \frac{d-2}{d} \quad (d \geq 3).$$

For $d \geq 2$, this equals the sharp Pham–Zakharov non-averaging exponent in dimension $d - 1$.

4. STRUCTURAL INPUT AND UNSHIFTED PREPROCESSING

We use the higher-dimensional homogeneous subset-sum structure theorem of Conlon–Fox–Pham [2] in the form recorded by Pham–Zakharov [5, Corollary 5]. We state only the consequences required below.

Throughout the technical argument, the ambient dimension and every displayed polynomial-box exponent are fixed. Parameters such as δ , γ and μ may depend on the set size n , and all $o(\cdot)$ relations and limits are taken as $n \rightarrow \infty$ along the family under consideration.

We use the following GAP conventions throughout. A symmetric homogeneous GAP is data

$$P = \left\{ \sum_{i=1}^e n_i q_i : |n_i| \leq N_i \right\}, \quad q_i \in \mathbb{Z}^\ell, \quad N_i \in \mathbb{Z}_{\geq 0}.$$

Its *dimension* is the number e of displayed steps, while its *rank* is $\dim_{\mathbb{R}} \text{span}_{\mathbb{R}}\{q_1, \dots, q_e\}$. It is *proper* if its coefficient map is injective on the displayed box. For $t \geq 0$ we set

$$tP = \left\{ \sum_{i=1}^e n_i q_i : |n_i| \leq \lfloor tN_i \rfloor \right\}.$$

If P is proper, ϕ_P denotes the inverse coefficient identification from P to its coordinate box. This is an identification on P , not in general an ambient homomorphism $\mathbb{Z}^\ell \rightarrow \mathbb{Z}^e$. For a real number $x \geq 3$ we also write

$$s(x) = \frac{x}{(\log x)^2};$$

thus $s(C) = s(|C|)$ below.

Theorem 4.1 (Conlon–Fox–Pham; Pham–Zakharov). *For every fixed ambient dimension ℓ and every $\beta > 1$ there are constants $c > 0$ and $D \in \mathbb{N}$ with the following property. Let $C \subseteq \mathbb{Z}^\ell$ have cardinality m and be contained in an axis-parallel box Q satisfying $|Q| \leq m^\beta$. For all sufficiently large m there are a set $\widehat{C} \subseteq C$ with*

$$|\widehat{C}| \geq m - c^{-1} \frac{m}{\log m},$$

a symmetric GAP

$$P(C) = \left\{ \sum_{i=1}^{d'} n_i q_i : |n_i| \leq N_i \right\}, \quad d' \leq D,$$

containing $\widehat{C} \cup \{0\}$, and a set $C_0 \subseteq \widehat{C}$ with $|C_0| \leq s(C)$ such that

$$q + cs(C)P(C) \subseteq \Sigma(C_0)$$

for some $q \in s(C)P(C)$. Moreover $cs(C)P(C)$ is proper.

The existence statement through properness is the cited corollary of Pham–Zakharov, derived from the Conlon–Fox–Pham structure theorem. Their symmetric convention is the centered display above, so $P(C)$ is homogeneous. The location $q \in s(C)P(C)$ is not an extra external hypothesis: since $C_0 \subseteq \widehat{C} \subseteq P(C)$ and $|C_0| \leq s(C)$, one has $\Sigma(C_0) \subseteq s(C)P(C)$; since $0 \in cs(C)P(C)$, the core inclusion itself gives $q \in \Sigma(C_0) \subseteq s(C)P(C)$.

For fixed ambient dimension and polynomial-box exponent, a *structural package* means a fixed choice of the constants and of the objects $\widehat{C}, C_0, q, P(C)$ in Theorem 4.1 for every eligible sufficiently large set C . When dimensions vary over a fixed finite range, we choose one package in each ambient dimension before beginning an iteration and, after every coefficient identification, use the package indexed by the new ambient dimension; no post hoc choice of a polynomial-box exponent or of structural constants is ever made, so all constants are uniform once the finite dimension range and the common polynomial-box exponent have been fixed. Within each package we choose an admissible structural GAP of minimum dimension and write $d(C) = \dim P(C)$; ties are broken arbitrarily, and every radius then satisfies $N_i \geq 1$. For sufficiently large $|C|$, properness of $cs(C)P(C)$ implies properness of $P(C)$, and the centered representation makes $\phi_{P(C)}$ based at the origin. We suppress the chosen package from the notation.

Definition 4.2 (Unshifted irreducibility). Fix a structural package applicable to all sets under consideration. Let $V \subseteq B \subseteq \mathbb{Z}^d$, where B is a symmetric axis-parallel coordinate box. We say that the pair (V, B) is (δ, γ) -*irreducible in the unshifted sense*, relative to this package, if every $D \subseteq V$ with $|D| \geq \delta|V|$ has subset-sum dimension d and satisfies

$$|P(D)| \geq \gamma|B|.$$

The omission of translations is essential: shell non-divisibility is inherited by subsets and by coefficient identifications of proper homogeneous GAPs, with the shell functional transported as below, but not by arbitrary translations.

Lemma 4.3 (Transformation of the shell functional). *Let $P \subseteq \mathbb{Z}^\ell$ be a proper homogeneous GAP with step vectors $q_1, \dots, q_d$, and let $\lambda : \mathbb{Z}^\ell \rightarrow \mathbb{Z}$ be integer-linear. Define*

$$\lambda_P(n_1, \dots, n_d) = \sum_{i=1}^d n_i \lambda(q_i).$$

Then

$$\lambda_P(\phi_P(v)) = \lambda(v) \quad (v \in P).$$

Consequently, if $V \subseteq P$ is shell non-dividing for λ , then $\phi_P(V)$ is shell non-dividing for λ_P with exactly the same shell parameter.

Proof. Homogeneity gives $v = \sum_i n_i q_i$ whenever $\phi_P(v) = (n_1, \dots, n_d)$. The identity follows by linearity. Every prohibited divisibility relation is an identity involving only the integer values of the functional, so it is preserved exactly. $\square$

Lemma 4.4 (Progression comparison in a coordinate box). *Fix r and $\beta > 1$. Let $Q \subseteq \mathbb{Z}^r$ be a proper symmetric coordinate GAP and let $C \subseteq Q$ satisfy $|Q| \leq |C|^\beta$. Write $e = d(C)$ and $s = s(C)$, using the structural package for (r, β) . Then*

$$|P(C)| \ll_{r,\beta} s^{-(e-r)} |Q| \quad (e > r),$$

while

$$|P(C)| \ll_{r,\beta} |Q| \quad (e \leq r).$$

The constants are uniform when r, β and the structural dimensions under consideration range over fixed bounded sets.

Proof. Let $R = P(C)$, and let $C_0 \subseteq C$ be the core set supplied by Theorem 4.1. Since $C_0 \subseteq Q$ and $|C_0| \leq s$,

$$q + csR \subseteq \Sigma(C_0) \subseteq sQ.$$

If $e > r$, properness of csR and of the coordinate GAP Q gives

$$s^e |R| \ll |csR| \leq |sQ| \ll_r s^r |Q|,$$

which proves the first assertion.

Suppose $e \leq r$. Let H be the real span of the steps of R , of dimension $h \leq e$; for all sufficiently large $|C|$ the affine hull of $q + csR$ has direction space H . Differences of points of $q + csR \subseteq \Sigma(C_0) \subseteq \text{span}_{\mathbb{R}}(C_0)$ show that $H \subseteq \text{span}_{\mathbb{R}}(C_0)$; conversely $C_0 \subseteq R \subseteq H$, so C_0 spans H and, in particular, the lattice points of $H \cap \text{conv}(Q)$ span H . Apply the fixed-dimensional discrete John lemma [5, Lemma 7] to the symmetric convex body $H \cap \text{conv}(Q)$ in the lattice $H \cap \mathbb{Z}^r$, identified with $\mathbb{Z}^h$. Since Q is a coordinate box, $(H \cap \text{conv}(Q)) \cap \mathbb{Z}^r = H \cap Q$, so the lemma gives a proper symmetric rank- h GAP Q_H with

$$H \cap Q \subseteq Q_H, \quad |Q_H| \ll_r |H \cap Q| \leq |Q|.$$

The set C_0 lies in $R \cap Q \subseteq H \cap Q$, hence $q + csR \subseteq \Sigma(C_0) \subseteq sQ_H$, and consequently

$$s^e |R| \ll |csR| \leq |sQ_H| \ll_r s^h |Q_H| \ll_r s^e |Q|,$$

proving the second assertion. $\square$

Lemma 4.5 (Unshifted preprocessing). *Fix an ambient dimension ℓ , $\beta > 1$, $\varepsilon \in (0, 1/3)$ and a desired shrink exponent $K_0 \geq 1$. Let $V \subseteq B \subseteq \mathbb{Z}^\ell$ be shell non-dividing, with $|B| \leq |V|^\beta$. Suppose $0 < \delta \leq 1/2$ and*

$$\gamma \leq \delta^K, \quad \gamma \geq |V|^{-1/3},$$

where $K = K(\ell, \beta, \varepsilon, K_0)$ is sufficiently large. Then one can pass, using only subsets and coefficient identifications of proper homogeneous GAPs, to a shell non-dividing set $\tilde{V} \subseteq \tilde{B} \subseteq \mathbb{Z}^{\tilde{d}}$ and fix a structural package in $\mathbb{Z}^{\tilde{d}}$ such that $(\tilde{V}, \tilde{B})$ is (δ, γ) -irreducible relative to that package and

$$\begin{aligned} |\tilde{B}| &\ll_{\ell, \beta, \varepsilon, K_0} |V|^{-(1-\varepsilon)(\tilde{d}-\ell)} |B|, & |\tilde{V}| &> |V|^{1-\varepsilon}, & \tilde{d} &> \ell, \\ |\tilde{B}| &\ll_{\ell, \beta, \varepsilon, K_0} \rho^{K_0} |B|, & |\tilde{V}| &= \rho |V| > |V|^{1-\varepsilon}, & \tilde{d} &= \ell, \\ |\tilde{B}| &\ll_{\ell, \beta, \varepsilon, K_0} |B|, & |\tilde{V}| &> |V|^{1-\varepsilon}, & \tilde{d} &< \ell. \end{aligned}$$

Here $\tilde{d} = O_{\ell, \beta}(1)$; this dimension bound is uniform for $\varepsilon \in (0, 1/3)$ and $K_0 \geq 1$. The polynomial-box exponent and constants of the output structural package are bounded in $\ell, \beta, \varepsilon, K_0$.

Proof. Put $n = |V|$ and $\delta_* = \delta/2$; we assume throughout that n is sufficiently large in the fixed parameters and enlarge K whenever needed. Let $D_{\text{init}} = D(\ell, \beta)$ be the structural-dimension bound in Theorem 4.1, and choose

$$C_U > 3(\beta + 2), \quad \bar{d} > D_{\text{init}} + C_U, \quad \bar{\beta} = 4(\beta + 3).$$

Before beginning the iteration, fix the initial package for (ℓ, β) , structural packages with common polynomial-box exponent $\bar{\beta}$ in every ambient dimension $1 \leq r \leq \bar{d}$, the bound $D_* = \max_{1 \leq r \leq \bar{d}} D(r, \bar{\beta})$, and a constant $C = C(\bar{d}, \bar{\beta})$ dominating the constants of Lemma 4.4 whenever the ambient dimension is at most $\bar{d}$ and the output structural dimension is at most D_* .

Apply the structural package for (ℓ, β) to $V \subseteq B$, and write $A_0 = V$, $P_0 = P(V)$ and $d_0 = \dim P_0$. Lemma 4.4, applied initially in B , and $s(V) \geq n^{1-\varepsilon}$ give

$$|P_0| \ll_{\ell, \beta} n^{-(1-\varepsilon) \max(0, d_0 - \ell)} |B| \leq n^{\beta+1} \quad (1)$$

for all sufficiently large n .

At an accepted stage A_j let P_j be its assigned structural progression of dimension d_j , and define

$$A_j^\sharp = \phi_{P_j}(\hat{A}_j), \quad Q_j = \phi_{P_j}(P_j),$$

so that Q_j is a proper symmetric coordinate box of dimension d_j and cardinality $|P_j|$, while $|A_j^\sharp| = (1 - o(1))|A_j| \geq \frac{1}{2}|A_j|$. We impose the provisional eligibility condition

$$|Q_j| \leq (\delta|A_j^\sharp|)^{\bar{\beta}}, \quad (2)$$

if it fails the procedure stops, and if it holds every $D \subseteq A_j^\sharp$ with $|D| \geq \delta|A_j^\sharp|$ is eligible for the already fixed package in ambient dimension d_j .

If $(A_j^\sharp, Q_j)$ is not unshiftedly (δ, γ) -irreducible, choose a witnessing set $A_{j+1} \subseteq A_j^\sharp$ with $|A_{j+1}| \geq \delta|A_j^\sharp| \geq \delta_*|A_j|$, its structural data taken from the fixed package in dimension d_j . We call the proposed move a down-move if $d_{j+1} < d_j$, an up-move if $d_{j+1} > d_j$, and a shrink-move if $d_{j+1} = d_j$ and $|P_{j+1}| < \gamma|P_j|$. Lemma 4.4 gives

$$\begin{aligned} |P_{j+1}| &\leq C|P_j| && \text{for a down-move,} \\ |P_{j+1}| &\leq C s(A_{j+1})^{-(d_{j+1}-d_j)} |P_j| && \text{for an up-move,} \\ |P_{j+1}| &< \gamma|P_j| && \text{for a shrink-move.} \end{aligned}$$

At each coefficient identification the shell functional is transported by Lemma 4.3; hence every accepted set remains shell non-dividing.

Accept proposed moves sequentially, but stop before accepting one if its cardinality is at most $n^{1-\varepsilon/2}$ or if its structural dimension exceeds $\bar{d}$; otherwise stop when the eligibility condition fails or no move is available. Suppose that $A_0, \dots, A_i$ have been accepted, let $\mathcal{D}, \mathcal{U}, \mathcal{S}$ be the sets of down-, up- and shrink-move indices, and put $U = \sum_{j \in \mathcal{U}} (d_{j+1} - d_j)$. Multiplying the move estimates gives

$$|P_i| \leq C^{|\mathcal{D}|+|\mathcal{U}|} \gamma^{|\mathcal{S}|} \prod_{j \in \mathcal{U}} s(A_{j+1})^{-(d_{j+1}-d_j)} |P_0|, \quad (3)$$

while the dimension bookkeeping gives $d_i - d_0 \leq U - |\mathcal{D}|$, hence $|\mathcal{D}| \leq d_0 + U$ and $|\mathcal{U}| \leq U$; moreover $|A_i| \geq \delta_*^i n$. Every accepted set after A_0 has size greater than $n^{1-\varepsilon/2}$, so $s(A_j) \geq n^{1-\varepsilon}$

for every $j \geq 1$ occurring in an up-move. Since $|P_i| \geq 1$ and $\gamma \leq 1$, equation (3) together with (1) implies, once $C^{D_{\text{init}}} \leq n$ and $C^2 \leq n^{(1-\varepsilon)/2}$,

$$n^{(1-\varepsilon)U/2} \leq n^{\beta+2}, \quad \text{and hence} \quad U \leq \frac{2(\beta+2)}{1-\varepsilon} \leq 3(\beta+2) < C_U.$$

We check that none of the three provisional cutoffs ever triggers; the uniformity bookkeeping abbreviated here is carried out in full in the formalization (Section 1.2).

Dimension. Every accepted dimension is at most $d_0 + U < D_{\text{init}} + C_U < \bar{d}$. The identical calculation applies after adjoining any proposed move whose cardinality exceeds $n^{1-\varepsilon/2}$: its structural dimension is at most D_* , so the same constant C applies, the total upward change is still less than C_U , and the proposed dimension is also less than $\bar{d}$.

Eligibility. The same estimates applied to every partial sequence, discarding all factors at most one in (3), give

$$|P_i| \leq C^{D_{\text{init}}+2C_U} |P_0| \leq n^{\beta+2}. \quad (4)$$

On the other hand, $K \geq 1$, $\gamma \geq n^{-1/3}$ and $|A_i| > n^{1-\varepsilon/2}$ give $\delta|A_i^\sharp| \geq \frac{1}{2}n^{1-\varepsilon/2-1/(3K)} \geq \frac{1}{2}n^{2/3-\varepsilon/2}$; since $\varepsilon < 1/3$ and $\bar{\beta} = 4(\beta+3)$, the right-hand side raised to the power $\bar{\beta}$ exceeds $n^{\beta+2}$. Together with (4), this proves (2) at every accepted stage, so the entire argument uses only the packages fixed before the iteration.

Cardinality. Equation (3) and the bound on U give

$$|P_i| \ll_{\ell,\beta,\varepsilon} \gamma^{i-|\mathcal{U}|-|\mathcal{D}|} n^{-(1-\varepsilon)U} |P_0|. \quad (5)$$

A proposed set A_{i+1} of size at most $n^{1-\varepsilon/2}$ would satisfy $\delta_*^{i+1}n \leq |A_{i+1}| \leq n^{1-\varepsilon/2}$, hence $\delta_*^{i+1} \leq n^{-\varepsilon/2}$. Since $\delta \leq 1/2$, one has $\delta^2 \leq \delta_*$ and therefore $\gamma \leq \delta^K \leq \delta_*^{K/2}$. With $C_1 = C_1(\ell, \beta, \varepsilon)$ bounding $|\mathcal{U}| + |\mathcal{D}|$, $|P_0| \leq n^{\beta+1}$ and $\gamma \geq n^{-1/3}$, we get

$$\gamma^{i-|\mathcal{U}|-|\mathcal{D}|} |P_0| \leq \gamma^{i+1} \gamma^{-(C_1+1)} |P_0| \leq n^{-\varepsilon K/4+(C_1+1)/3+\beta+1}.$$

Taking K sufficiently large in $\ell, \beta, \varepsilon, K_0$ and then n sufficiently large makes this smaller than the reciprocal of the implicit constant in (5), contradicting $|P_i| \geq 1$.

The procedure therefore stops only because no move is available: every $D \subseteq A_i^\sharp$ with $|D| \geq \delta|A_i^\sharp|$ has $d(D) = d_i$ and $|P(D)| \geq \gamma|P_i|$, so $(A_i^\sharp, Q_i)$ is unshiftedly (δ, γ) -irreducible relative to the fixed package in dimension d_i . Put

$$\tilde{V} = A_i^\sharp, \quad \tilde{B} = Q_i, \quad \tilde{d} = d_i.$$

The deletion estimate and the stopping rule give $|\tilde{V}| > |V|^{1-\varepsilon}$, and the terminal package belongs to the preselected finite collection, whose dimensions, exponent $\bar{\beta}$, constants and thresholds depend only on $\ell, \beta, \varepsilon, K_0$.

It remains to prove the three box estimates. Combining (5) with (1) yields

$$|P_i| \ll_{\ell,\beta,\varepsilon} \gamma^{i-|\mathcal{U}|-|\mathcal{D}|} n^{-(1-\varepsilon)(U+\max(0,d_0-\ell))} |B|. \quad (6)$$

If $d_i > \ell$, then $U + \max(0, d_0 - \ell) \geq d_i - \ell$, so (6) gives the first displayed alternative; if $d_i < \ell$, discarding all factors at most one gives $|P_i| \ll_{\ell,\beta,\varepsilon} |B|$. Finally suppose $d_i = \ell$ and put $M = U + \max(0, d_0 - \ell)$, so that the dimension bookkeeping gives $|\mathcal{U}| + |\mathcal{D}| \leq 2M$. Since $\varepsilon < 1/3$ and $\gamma \geq n^{-1/3}$, one has $n^{-(1-\varepsilon)M} \leq \gamma^{2M}$, and equation (6) gives $|P_i| \ll_{\ell,\beta,\varepsilon} \gamma^i |B|$. Furthermore $|A_i| \geq \delta_*^i n$ and $|\tilde{V}| = (1 - o(1))|A_i|$, so $\rho = |\tilde{V}|/n$ satisfies $\delta_*^i \ll \rho$; since $\gamma \leq \delta_*^{K/2}$, choosing $K \geq 2K_0$ sufficiently large gives $\gamma^i \ll_{K_0} \rho^{K_0}$. This proves the same-dimensional alternative and completes the proof. $\square$

5. THREE GEOMETRIC-ARITHMETIC LEMMAS

We next establish the slab, rounding and lattice-index statements used in the tube argument.

5.1. A slab dichotomy. For a symmetric real box $B_{\mathbb{R}} = \prod_{i=1}^d [-N_i, N_i]$ with $N_i \in \mathbb{Z}_{\geq 1}$, write $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$ and

$$w_B(u) = \max_{x \in B_{\mathbb{R}}} |u(x)|.$$

Lemma 5.1 (Convex lattice sets and their John progression). *Let $H \subseteq \mathbb{R}^d$ be an r -dimensional rational subspace, let $\Lambda = H \cap \mathbb{Z}^d$, and let $K \subseteq H$ be an origin-symmetric convex body such that $K \cap \Lambda$ spans H . Then $K \cap \Lambda$ is contained in a proper symmetric rank- r GAP $Q \subseteq \Lambda$ satisfying*

$$|Q| \ll_d |K \cap \Lambda|.$$

If $r = d$, then additionally

$$|K \cap \mathbb{Z}^d| \ll_d \text{vol}_d(K).$$

Proof. The case $r = 0$ is immediate. For $r \geq 1$, identify Λ with $\mathbb{Z}^r$ and apply the discrete John lemma [5, Lemma 7]. It gives a lattice basis $v_1, \dots, v_r$, radii $N_i \geq 1$, and a constant $c_r > 0$ such that

$$\left\{ \sum n_i v_i : |n_i| \leq \lfloor c_r N_i \rfloor \right\} \subseteq K \cap \Lambda \subseteq Q := \left\{ \sum n_i v_i : |n_i| \leq N_i \right\}.$$

The outer progression is proper and $|Q| \ll_r |K \cap \Lambda|$. If $r = d$, the successive-minima lattice-point bound and Minkowski's second theorem give

$$|K \cap \mathbb{Z}^d| \ll_d \prod_{i=1}^d \lambda_i^{-1} \ll_d \text{vol}_d(K),$$

since the last successive minimum is at most one. $\square$

Lemma 5.2 (Lattice points in a central slab). *Let u be a nonzero real linear functional, let $0 < \tau \leq 1$, and put*

$$K = B_{\mathbb{R}} \cap \{x : |u(x)| \leq \tau w_B(u)\}, \quad S = K \cap \mathbb{Z}^d.$$

If the real span of S has dimension r , then S is contained in a proper symmetric rank- r GAP Q satisfying

$$|Q| \ll_d \begin{cases} \tau |B|, & r = d, \\ |B|, & r < d. \end{cases}$$

Proof. Let $H = \text{span}_{\mathbb{R}} S$ and $\Lambda = H \cap \mathbb{Z}^d$. The set S is precisely the lattice-point set of the origin-symmetric convex body $K \cap H$ in Λ , and it spans H . Lemma 5.1 therefore places S in a proper rank- r GAP Q with

$$|Q| \ll_d |S|.$$

If $r < d$, the trivial estimate $|S| \leq |B|$ proves the second alternative.

Assume $r = d$. Lemma 5.1 gives

$$|S| \ll_d \text{vol}_d(K).$$

Write $u(x) = \sum_i a_i x_i$ and choose an index j for which $|a_j| N_j \geq w_B(u)/d$. After fixing all coordinates other than x_j , the allowed values of x_j form an interval whose length is at most

$$\frac{2\tau w_B(u)}{|a_j|} \leq 2d\tau N_j.$$

Fubini's theorem consequently gives

$$\text{vol}_d(K) \leq d\tau \text{vol}_d(B_{\mathbb{R}}).$$

Since the side widths of the coordinate box are positive integers, $\text{vol}_d(B_{\mathbb{R}}) \asymp_d |B|$. Combining the last three displays proves $|Q| \ll_d \tau |B|$. $\square$

Lemma 5.3 (Slab exclusion from irreducibility). *Fix d and $\beta > 1$. Let $V \subseteq B \subseteq \mathbb{Z}^d$, and fix a structural package arising from Theorem 4.1 in ambient dimension d with polynomial-box exponent at most β , applicable to every subset of V of size at least $\delta |V|$. Suppose that (V, B) is unshiftedly (δ, γ) -irreducible relative to this package, and put $n = |V|$, and assume*

$$|B| \leq n^\beta, \quad \delta \geq n^{-1/2}, \quad [s(\delta n)]^{-1} = o(\gamma).$$

There is $c_{d,\beta} > 0$ such that every central slab of relative width at most $c_{d,\beta}\gamma$ contains fewer than δn points of V .

Proof. Suppose that $D \subseteq V$ lies in a central slab of relative width $\tau \leq c_{d,\beta}\gamma$ and that $|D| \geq \delta n$. Since $\delta \geq n^{-1/2}$, one has $|B| \leq n^\beta \leq |D|^{2\beta}$, so the fixed structural package is applicable to D , and irreducibility gives $d(D) = d$ and $|P(D)| \geq \gamma |B|$.

Let r be the dimension of the real span of all lattice points in the slab, let Q be supplied by Lemma 5.2, and put $s_D = s(D)$. The structural core $D_0 \subseteq D \subseteq Q$ has size at most s_D , so $q + cs_D P(D) \subseteq \Sigma(D_0) \subseteq s_D Q$, and the left-hand GAP is proper. Consequently

$$s_D^d |P(D)| \ll_{d,\beta} |cs_D P(D)| \leq |s_D Q| \ll_d s_D^r |Q|.$$

If $r < d$, then $|P(D)| \ll_{d,\beta} s_D^{r-d} |Q| \leq C_{d,\beta} s_D^{-1} |B|$; since $x/(\log x)^2$ is increasing for large x , one has $s_D \geq s(\delta n)$, and the parameter hypothesis makes the last expression $o(\gamma |B|)$, a contradiction. If $r = d$, then

$$|P(D)| \ll_{d,\beta} |Q| \ll_d \tau |B| \leq C_{d,\beta} c_{d,\beta} \gamma |B|,$$

and choosing $c_{d,\beta}$ sufficiently small gives the final contradiction. $\square$

5.2. One-sided filling around a zonotope. For a finite set $E \subseteq \mathbb{R}^d$, write

$$Z(E) = \left\{ \sum_{v \in E} t_v v : 0 \leq t_v \leq 1 \right\}$$

for its zonotope. For a homogeneous GAP P , let $\langle P \rangle$ be the lattice generated by its steps.

Lemma 5.4 (Rounding with a homogeneous core). *Let $C \subseteq B \subseteq \mathbb{Z}^d$, let $P = P(C)$ have dimension d , and assume that its step lattice*

$$L = \langle P \rangle$$

has rank d . Take the objects $\widehat{C}, C_0, q$ from Theorem 4.1, and put

$$\overline{C} = \widehat{C} \setminus C_0, \quad s = s(C).$$

For all sufficiently large $|C|$,

$$L \cap \left(Z(\overline{C}) + q + \frac{cs}{2} P \right) \subseteq \Sigma(C).$$

Proof. Write $P = \{\sum_{i=1}^d n_i q_i : |n_i| \leq N_i\}$. The rank assumption and the fact that P has exactly d steps imply that $q_1, \dots, q_d$ are linearly independent, hence a $\mathbb{Z}$ -basis of L , and the coefficient map extends to a real-linear isomorphism $\psi : \text{span}_{\mathbb{R}} L \rightarrow \mathbb{R}^d$, $\psi(\sum_i x_i q_i) = (x_1, \dots, x_d)$, with $\psi(P) = \prod_{i=1}^d [-N_i, N_i] \cap \mathbb{Z}^d$.

Let

$$y = z + t \in L \cap \left(Z(\overline{C}) + q + \frac{cs}{2} P \right),$$

where $z \in Z(\overline{C})$ and $t \in q + (cs/2)P$. Since $q \in sP$, both q and t belong to L ; hence $z = y - t \in L$ and $\psi(z) \in Z(\psi(\overline{C}))$. Apply the Conlon–Fox–Pham zonotope-rounding lemma in the form of [5, Lemma 13] to $A^* = \psi(\overline{C}) \cup \{0\}$ inside the coefficient box $\prod_i [-N_i, N_i] \cap \mathbb{Z}^d$, whose i th width is $2N_i + 1$ in the convention of that lemma. It gives a subset sum $\tilde{z}' \in \Sigma(\psi(\overline{C}))$ such that

$$|\psi(z)_i - \tilde{z}'_i| \leq \sqrt{d|A^*|} (2N_i + 1) \leq 3\sqrt{2d|C|} N_i \quad (1 \leq i \leq d),$$

using $|A^*| \leq |C| + 1 \leq 2|C|$ and $N_i \geq 1$; the rounding error carries the sign of [5, equation (15)], whose rounding is written $z = \tilde{z} + r$. Pulling back by ψ gives a subset sum $\tilde{z} \in \Sigma(\overline{C})$ with

$$z - \tilde{z} \in C_d \sqrt{|C|} P, \quad C_d = 3\sqrt{2d}.$$

Since $s = |C|/(\log |C|)^2$, one has $C_d \sqrt{|C|} \leq cs/2$ for all sufficiently large $|C|$. As $t \in q + (cs/2)P$, coefficientwise addition in the symmetric GAP therefore gives

$$(z - \tilde{z}) + t \in q + csP \subseteq \Sigma(C_0).$$

The subset producing $\tilde{z}$ lies in $\overline{C}$ and the subset producing $(z - \tilde{z}) + t$ lies in C_0 ; these are disjoint, so $y = \tilde{z} + ((z - \tilde{z}) + t) \in \Sigma(C)$. $\square$

5.3. A large progression has small lattice index.

Lemma 5.5 (Containment of the structural GAP). *Let $C \subseteq B \subseteq \mathbb{Z}^d$, where $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$ and $B_{\mathbb{R}}$ is a symmetric axis-parallel real box. Let $P = P(C)$ and let c be the constant in Theorem 4.1. For all sufficiently large $|C|$,*

$$P \subseteq \frac{4}{c} B_{\mathbb{R}}.$$

In particular, when the ambient dimension and polynomial-box exponent are fixed, $P \subseteq C_{d,\beta} B_{\mathbb{R}}$.

Proof. Let $s = s(C)$ and let C_0, q be the structural core and translation. Since $C_0 \subseteq B$ and $|C_0| \leq s$,

$$q + csP \subseteq \Sigma(C_0) \subseteq sB_{\mathbb{R}}.$$

As $0 \in csP$, also $q \in sB_{\mathbb{R}}$, and hence $csP \subseteq 2sB_{\mathbb{R}}$. For $p \in P$, put $k = \lfloor cs \rfloor \geq cs/2$. Then $kp \in csP$, so $p \in (2s/k)B_{\mathbb{R}} \subseteq (4/c)B_{\mathbb{R}}$. $\square$

Lemma 5.6 (Lattice sections of a coordinate box). *Let*

$$B_{\mathbb{R}} = \prod_{i=1}^d [-M_i, M_i], \quad B = B_{\mathbb{R}} \cap \mathbb{Z}^d,$$

where the M_i are positive integers. If $H \subseteq \mathbb{R}^d$ is an r -dimensional linear subspace and $t \geq 1$, then

$$|H \cap tB_{\mathbb{R}} \cap \mathbb{Z}^d| \leq t^r |B|.$$

Proof. Choose $I \subseteq [d]$, $|I| = r$, so that the coordinate projection $\text{pr}_I : H \rightarrow \mathbb{R}^I$ is injective. It is then injective on $H \cap \mathbb{Z}^d$, and therefore

$$|H \cap tB_{\mathbb{R}} \cap \mathbb{Z}^d| \leq \prod_{i \in I} (2\lfloor tM_i \rfloor + 1) \leq t^r \prod_{i \in I} (2M_i + 1) \leq t^r |B|.$$

$\square$

Lemma 5.7 (Full rank and index). *Fix d and $\beta > 1$, and fix a structural package arising from Theorem 4.1 in ambient dimension d with polynomial-box exponent at most β . Let (V, B) be unshiftedly (δ, γ) -irreducible relative to this package, where $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$ is a symmetric coordinate box. Let $C \subseteq V$ have $|C| \geq \delta|V|$ and assume $|B| \leq |C|^\beta$. Put*

$$P = P(C), \quad s = s(C), \quad L = \langle P \rangle.$$

If

$$s\gamma \longrightarrow \infty,$$

then L has rank d and

$$[\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}.$$

Proof. Irreducibility gives $\dim P = d$ and $|P| \geq \gamma|B|$. Let $c = c(d, \beta) > 0$ be the structural constant in Theorem 4.1; the proof of Lemma 5.5 gives the stronger core containment

$$csP \subseteq 2sB_{\mathbb{R}}. \quad (7)$$

Write $P = \{\sum_{i=1}^d n_i q_i : |n_i| \leq N_i\}$, where every $N_i \geq 1$ as zero-width directions are not counted in the dimension, and let $H = \text{span}_{\mathbb{R}}\{q_1, \dots, q_d\}$, of dimension r . Because csP is proper, its cardinality is the product of its coefficient widths, so for all sufficiently large s

$$|csP| = \prod_{i=1}^d (2\lfloor csN_i \rfloor + 1) \gg_{d,\beta} s^d |P|, \quad (8)$$

while (7) and Lemma 5.6 give

$$|csP| \leq |H \cap 2sB_{\mathbb{R}} \cap \mathbb{Z}^d| \leq (2s)^r |B|. \quad (9)$$

Combining (8) and (9) with $|P| \geq \gamma|B|$ yields $s^{d-r}\gamma \ll_{d,\beta} 1$. If $r < d$, the left-hand side is at least $s\gamma$ for large s , contradicting the hypothesis. Hence $r = d$, the d step vectors are linearly independent, and they form a $\mathbb{Z}$ -basis of their integer span L .

Let M be the $d \times d$ matrix whose columns are $q_1, \dots, q_d$, so that $L = M\mathbb{Z}^d$ and $[\mathbb{Z}^d : L] = |\det M|$. Since $\text{conv } P = M(\prod_{i=1}^d [-N_i, N_i])$,

$$\text{vol}_d(\text{conv } P) = 2^d |\det M| \prod_{i=1}^d N_i \asymp_d [\mathbb{Z}^d : L] |P|, \quad (10)$$

the comparison using $N_i \geq 1$ and therefore uniform in the widths. Lemma 5.5 and convexity give $\text{conv } P \subseteq C_{d,\beta} B_{\mathbb{R}}$, and $\text{vol}_d(B_{\mathbb{R}}) \asymp_d |B|$, so equation (10) implies $[\mathbb{Z}^d : L] |P| \ll_{d,\beta} |B|$. Using $|P| \geq \gamma|B|$ now gives $[\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}$, as claimed. $\square$

Lemma 5.8 (Full-dimensional projective support). *Fix d and $\beta > 1$, and fix a structural package arising from Theorem 4.1 in ambient dimension d with polynomial-box exponent at most β . Let $V \subseteq B \subseteq \mathbb{Z}^d$ be (λ, X) -shell non-dividing and unshiftedly (δ, γ) -irreducible relative to this package, put $n = |V|$, and assume*

$$|B| \leq n^\beta, \quad [s(\delta n)]^{-1} = o(\gamma).$$

Then, for all sufficiently large n ,

$$\text{span}_{\mathbb{R}} V = \mathbb{R}^d, \quad \text{aff } \pi(V) = H_\lambda.$$

Moreover

$$\pi(V) \subseteq \Omega := \{y \in H_\lambda : Xy \in B_{\mathbb{R}}\},$$

and Ω has nonempty relative interior in H_λ .

Proof. Apply Lemma 5.7 with $C = V$. Since $s(V) \geq s(\delta n)$ for all sufficiently large n , the hypothesis implies $s(V)\gamma \rightarrow \infty$. The structural step lattice $L = \langle P(V) \rangle$ therefore has rank d .

Let $V_0 \subseteq V$ and q be the structural core and translation from Theorem 4.1. For sufficiently large n , every coordinate radius of $cs(V)P(V)$ is at least one. Since

$$q + cs(V)P(V) \subseteq \Sigma(V_0) \subseteq \text{span}_{\mathbb{R}} V,$$

taking differences of points in the left-hand translate shows that every step of $P(V)$ belongs to $\text{span}_{\mathbb{R}} V$. These steps have rank d , so $\text{span}_{\mathbb{R}} V = \mathbb{R}^d$.

If $\pi(V)$ were contained in a proper affine subspace of H_λ , then its linear span would have dimension at most $d - 1$. Since $v = \lambda(v)\pi(v)$ for every $v \in V$, this would put V in a proper linear subspace of $\mathbb{R}^d$, a contradiction. Hence $\text{aff } \pi(V) = H_\lambda$.

Finally, for $v \in V$ the shell condition gives $X/\lambda(v) \in [1/2, 1]$. As $B_{\mathbb{R}}$ is convex, symmetric and contains the origin,

$$X\pi(v) = \frac{X}{\lambda(v)}v \in B_{\mathbb{R}}.$$

Thus $\pi(V) \subseteq \Omega$. Since Ω is convex and contains an affinely spanning subset of H_λ , it has nonempty relative interior. $\square$

6. PROJECTIVE CONVEXITY FROM A TUBE OF SUBSET SUMS

We now prove the central projective statement.

Definition 6.1 (Convex position for an indexed multiset). Let E be a finite-dimensional Euclidean affine space. An indexed family $p_1, \dots, p_n \in E$ is in μ -convex position if, for every index j , there is a closed half-space of E containing p_j and containing at most μn indices, counted with multiplicity. In a zero-dimensional affine space the only nonempty half-space is the whole space.

Lemma 6.2 (Projective duality). *Let V be a finite set lying in a (λ, X) -shell, and suppose that the indexed projective family $\pi(V)$ is not in μ -convex position. If $\mu n \geq 4$, then there are $a \in V$ and coefficients β_v for $v \in V \setminus \{a\}$ such that*

$$a = \sum_{v \neq a} \beta_v v, \quad 0 \leq \beta_v \leq \frac{4}{\mu n}, \quad \frac{1}{3} \leq \sum_{v \neq a} \beta_v \leq 3.$$

Proof. Choose an index a for which the convex-position condition fails. Consider the compact convex set

$$\mathcal{K} = \left\{ \sum_{v \in V} c_v (\pi(v) - \pi(a)) : 0 \leq c_v \leq (\mu n)^{-1}, \sum_v c_v = 1 \right\}.$$

We claim that $0 \in \mathcal{K}$. Otherwise a separating functional u would be negative on all of $\mathcal{K}$. But failure of convex position says that the half-space

$$\{x : u(x - \pi(a)) \geq 0\}$$

contains more than μn indices. Assigning total mass one, with each coefficient at most $(\mu n)^{-1}$, to those indices gives an element of $\mathcal{K}$ on which u is nonnegative, a contradiction.

Thus

$$\pi(a) = \sum_v c_v \pi(v).$$

Multiplying by $\lambda(a)$ gives

$$a = \sum_v c_v \frac{\lambda(a)}{\lambda(v)} v.$$

The shell condition puts every ratio $\lambda(a)/\lambda(v)$ in $[1/2, 2]$. Move the possible $v = a$ term to the left and divide by $1 - c_a$. Since $c_a \leq (\mu n)^{-1} \leq 1/4$, the stated bounds follow. $\square$

Lemma 6.3 (Robust weighted zonotope). *Fix d and $\beta > 1$. Let $V \subseteq B \subseteq \mathbb{Z}^d$, and fix a structural package arising from Theorem 4.1 in ambient dimension d with polynomial-box exponent at most β , applicable to every subset of V of size at least $\delta|V|$. Suppose that (V, B) is unshiftedly (δ, γ) -irreducible relative to this package, put $n = |V|$, and assume*

$$|B| \leq n^\beta, \quad \delta \geq n^{-1/2}, \quad \delta = o(\mu), \quad [s(\delta n)]^{-1} = o(\gamma).$$

Let $a \in V$ and let weights $(\theta_v)_{v \in V \setminus \{a\}}$ satisfy

$$0 \leq \theta_v \leq \frac{1}{2}, \quad c_1 \mu n \leq \sum_{v \neq a} \theta_v \leq c_2 \mu n$$

for fixed positive constants c_1, c_2 . Let $R \subseteq V \setminus \{a\}$ have size $o(\mu n)$, put $C = (V \setminus \{a\}) \setminus R$, and set

$$z_C = \sum_{v \in C} \theta_v v.$$

Then, for a constant $c_{d,\beta,c_1,c_2} > 0$,

$$z_C + c_{d,\beta,c_1,c_2} \mu \gamma n B_{\mathbb{R}} \subseteq Z(C).$$

Proof. Let

$$Z^0 = \sum_{v \in C} [-\theta_v, \theta_v] v.$$

It suffices to prove $\xi n B_{\mathbb{R}} \subseteq Z^0$ with $\xi = c_{d,\beta,c_1,c_2} \mu \gamma$.

Suppose $y \in \xi n B_{\mathbb{R}} \setminus Z^0$. By separation, after changing the sign of the separator if necessary, there is a linear functional u such that

$$u(y) > \sum_{v \in C} \theta_v |u(v)|.$$

Put $z = u(y) > 0$ and

$$H = \left\{ x : |u(x)| < \frac{c'_d \gamma z}{\xi n} \right\}.$$

Since $y \in \xi n B_{\mathbb{R}}$, this is a central slab of relative width at most $c'_d \gamma$. Lemma 5.3 therefore gives $|C \cap H| < \delta n$.

The total coefficient mass is $\asymp \mu n$. The removal of R costs $o(\mu n)$, and the mass on H is at most $\delta n/2 = o(\mu n)$. Hence

$$\sum_{v \in C \setminus H} \theta_v \gg \mu n.$$

For $v \notin H$,

$$|u(v)| \geq \frac{c'_d \gamma z}{\xi n}.$$

Thus

$$\sum_{v \in C} \theta_v |u(v)| \gg \mu n \frac{\gamma z}{\xi n} > z$$

when c_{d,β,c_1,c_2} is sufficiently small, contradicting the separating inequality. $\square$

Theorem 6.4 (Projective convexity). *Fix d and $\beta > 1$. Let $V \subseteq B \subseteq \mathbb{Z}^d$ be an (λ, X) -shell non-dividing set of size n , where B is a symmetric coordinate box and $|B| \leq n^\beta$, and fix a structural package arising from Theorem 4.1 in ambient dimension d with polynomial-box exponent at most β , applicable to every subset of V of size at least $\delta|V|$. Suppose that (V, B) is unshiftedly (δ, γ) -irreducible relative to this package. Assume $0 < \delta, \gamma, \mu < 1$ and the parameter hierarchy*

$$\delta \geq n^{-1/2}, \quad \delta = o(\mu), \quad \frac{n}{\log n} = o(\mu\gamma n), \quad \gamma^{-1} = o(\mu\gamma n), \quad [s(\delta n)]^{-1} = o(\gamma).$$

Then the indexed family $\pi(V)$ is in μ -convex position for all sufficiently large n .

Proof. Suppose otherwise, and use Lemma 6.2 to obtain a and coefficients β_v . Choose a sufficiently small absolute constant $c_0 > 0$ and set

$$T = \lfloor c_0 \mu n \rfloor, \quad \theta_v = T \beta_v.$$

Then

$$Ta = \sum_{v \neq a} \theta_v v, \quad 0 \leq \theta_v \leq \frac{1}{2}, \quad \sum_{v \neq a} \theta_v \asymp \mu n.$$

Put $C = V \setminus \{a\}$ and let $\widehat{C}, C_0, P, q$ be the objects assigned to C by the fixed structural package. Put

$$L = \langle P \rangle, \quad \overline{C} = \widehat{C} \setminus C_0,$$

and write $s = s(C)$. For sufficiently large n , one has $|C| \geq \delta n$. Irreducibility therefore gives $d(C) = d$ and $|P| \geq \gamma|B|$. Moreover, the hypothesis $[s(\delta n)]^{-1} = o(\gamma)$ implies $s(C)\gamma \rightarrow \infty$. Since $|C| = n - 1$, the bound $|B| \leq n^\beta$ implies $|B| \leq |C|^{2\beta}$ for sufficiently large n . Lemma 5.7, with polynomial-box exponent 2β , consequently applies before any rounding is performed and gives

$$\text{rank } L = d, \quad [\mathbb{Z}^d : L] \ll_{d,\beta} \gamma^{-1}.$$

The discarded set $R = C \setminus \overline{C}$ has size $O(n/\log n) = o(\mu n)$, so Lemma 6.3 gives

$$\bar{z} + \xi_0 n B_{\mathbb{R}} \subseteq Z(\overline{C}), \quad \bar{z} = \sum_{v \in \overline{C}} \theta_v v, \quad \xi_0 \asymp_{d,\beta} \mu \gamma,$$

while $Ta - \bar{z} \in O(n/\log n)B_{\mathbb{R}}$. Lemma 5.5 gives $P \subseteq C_{d,\beta}B_{\mathbb{R}}$, and $q \in sP$; hence

$$(Ta - \bar{z}) - q \in O\left(\frac{n}{\log n} + s\right)B_{\mathbb{R}} = o(\xi_0 n)B_{\mathbb{R}},$$

so the choice $\xi = \xi_0/2$ gives, for all sufficiently large n ,

$$(Ta - \bar{z}) - q + \xi n B_{\mathbb{R}} \subseteq \xi_0 n B_{\mathbb{R}}.$$

Consequently, if $y \in L \cap (Ta + \xi n B_{\mathbb{R}})$, then $y - q - \bar{z} \in \xi_0 n B_{\mathbb{R}}$, so $y - q \in Z(\overline{C})$ and $y \in Z(\overline{C}) + q + \frac{cs}{2}P$; Lemma 5.4 therefore gives the tube inclusion

$$\boxed{L \cap (Ta + \xi n B_{\mathbb{R}}) \subseteq \Sigma(C)}. \quad (\dagger)$$

Let r be the order of $a + L$ in $\mathbb{Z}^d/L$. Then $r \ll_{d,\beta} \gamma^{-1}$ and $ra \in L$. Since $r = o(\xi n)$ and $r = o(T)$, the nearest multiple of r to T is positive; call it m_0 . It satisfies

$$|m_0 - T| \leq \frac{r}{2} < \xi n.$$

As $a \in B$,

$$m_0 a \in L \cap (Ta + \xi n B_{\mathbb{R}}).$$

The tube inclusion gives $m_0 a \in \Sigma(V \setminus \{a\})$. Applying λ yields

$$m_0 \lambda(a) = \sum_{v \in S} \lambda(v)$$

for a nonempty $S \subseteq V \setminus \{a\}$, contradicting shell non-divisibility. $\square$

7. WEIGHTED PROJECTIVE DENSITY INCREMENT

We require an indexed version of the convex-density lemma. Rather than retranscribing the geometric proof, we derive it from the published set version by a perturbation and compactness argument.

Lemma 7.1 (Weighted convex-density lemma). *Fix $r \geq 1$ and $\varepsilon > 0$. There are $\tau \in (0, 1)$ and $\mu_0 > 0$ such that the following holds. Let $p_1, \dots, p_n$ be an indexed family in a convex body $\Omega \subseteq \mathbb{R}^r$ with nonempty interior, in μ -convex position, where $0 < \mu < \mu_0$, and assume that n is sufficiently large in r, ε and μ . Then there are*

$$\eta \in [\mu, \mu^\tau]$$

and a compact convex set $\Omega' \subseteq \Omega$ such that

$$\text{vol}_r(\Omega') \leq \eta \text{vol}_r(\Omega)$$

and

$$\#\{i : p_i \in \Omega'\} \geq \eta^{\kappa_r + \varepsilon} n, \quad \kappa_r = \frac{r-1}{r+1}.$$

For $r = 1$ this means $\kappa_1 = 0$.

Proof. Take τ and μ_0 from [5, Lemma 1]. For each j , choose a witnessing closed half-space H_j containing p_j and at most μn indices. For each $m \geq 1$, perturb the points successively to distinct points $q_i^{(m)} \in H_i \cap B(p_i, 1/m)$ chosen so close to p_i that $q_i^{(m)} \in H_j$ only when $p_i \in H_j$; this is possible because only finitely many of the half-spaces H_j omit p_i , each at positive distance from it, and $H_i \cap B(p_i, 1/m)$ is infinite since $r \geq 1$. Then each H_j contains $q_j^{(m)}$ and at most as many perturbed points as original indices, so the set $Q_m = \{q_i^{(m)} : 1 \leq i \leq n\}$ is in μ -convex position and lies in $\Omega_m = \Omega + m^{-1}\overline{B}_2^r$.

The set version of the Pham–Zakharov lemma gives $\eta_m \in [\mu, \mu^\tau]$ and a convex set $K_m \subseteq \Omega_m$, which we may take compact by passing to its closure, with

$$\text{vol}_r(K_m) \leq \eta_m \text{vol}_r(\Omega_m), \quad |Q_m \cap K_m| \geq \eta_m^{\kappa_r + \varepsilon} n.$$

The bodies K_m lie in the fixed bounded set $\Omega + \overline{B}_2^r$. Pass to a subsequence on which $\eta_m \rightarrow \eta$, the finite index set $\{i : q_i^{(m)} \in K_m\}$ is constant, and K_m converges in the Hausdorff metric to a compact convex set K ; Blaschke selection and the continuity of convex-body volume under Hausdorff convergence may be found in [6]. Since Ω_m decreases to Ω , one has $K \subseteq \Omega$, and every stabilized index yields $p_i \in K$. Consequently

$$\text{vol}_r(K) \leq \eta \text{vol}_r(\Omega), \quad \#\{i : p_i \in K\} \geq \eta^{\kappa_r + \varepsilon} n,$$

and $\Omega' = K$ has the required properties. $\square$

We now lift a projective cap back to the lattice.

Lemma 7.2 (Projective cap lifting). *Let $B_{\mathbb{R}} \subseteq \mathbb{R}^d$ be a symmetric coordinate box, let $B = B_{\mathbb{R}} \cap \mathbb{Z}^d$, and let $\lambda : \mathbb{R}^d \rightarrow \mathbb{R}$ be a nonzero integer linear functional. Equip*

$$H_\lambda = \{y \in \mathbb{R}^d : \lambda(y) = 1\}$$

with its induced Euclidean $(d-1)$ -dimensional measure m_λ , and put

$$\Omega = \{y \in H_\lambda : Xy \in B_{\mathbb{R}}\}.$$

Let $\Omega' \subseteq \Omega$ be compact and convex with

$$m_\lambda(\Omega') \leq \eta m_\lambda(\Omega).$$

Finally, let

$$W = \{v \in B : X \leq \lambda(v) \leq 2X, \pi(v) \in \Omega'\}.$$

Then there are a symmetric coordinate box $B' \subseteq \mathbb{Z}^r$, an integer-linear functional $\lambda' : \mathbb{Z}^r \rightarrow \mathbb{Z}$, and an injective map $\phi : W \rightarrow B'$ such that

$$\lambda'(\phi(v)) = \lambda(v) \quad (v \in W),$$

and one of the following holds:

- (i) $r < d$ and $|B'| \ll_d |B|$;
- (ii) $r = d$ and $|B'| \ll_d \eta |B|$.

In particular, shell non-divisibility and the shell parameter are preserved exactly under the identification.

Proof. Write $\lambda(x) = \ell \cdot x$ for a nonzero vector $\ell \in \mathbb{Z}^d$. For a measurable set $E \subseteq H_\lambda$ and $T > 0$, put $\text{Cone}_T(E) = \{ty : y \in E, 0 \leq t \leq T\}$. Integrating the Jacobian $t^{d-1}/\|\ell\|_2$ on $H_\lambda = \ell/\|\ell\|_2^2 + \ker \lambda$ gives

$$\text{vol}_d(\text{Cone}_T(E)) = \frac{T^d}{d\|\ell\|_2} m_\lambda(E), \quad (11)$$

so the factor depending on λ cancels from volume ratios.

Define $K = B_\mathbb{R} \cap \text{Cone}_{2X}(\Omega')$; it is convex, contains 0, and contains W . Since $\text{Cone}_X(\Omega) \subseteq B_\mathbb{R}$ (as $Xy \in B_\mathbb{R}$ for $y \in \Omega$ and $B_\mathbb{R}$ is convex and contains the origin), formula (11) gives

$$\begin{aligned} \text{vol}_d(K) &\leq \text{vol}_d(\text{Cone}_{2X}(\Omega')) \\ &\leq 2^d \eta \text{vol}_d(\text{Cone}_X(\Omega)) \leq 2^d \eta \text{vol}_d(B_\mathbb{R}). \end{aligned} \quad (12)$$

Put $S = (K - K) \cap \mathbb{Z}^d$, $H = \text{span}_\mathbb{R} S$ and $r = \dim H$. Since $0 \in K$ and $W \subseteq K \cap \mathbb{Z}^d$, one has $W \subseteq S$; also $K - K \subseteq 2B_\mathbb{R}$. Applying Lemma 5.1 in the lattice $H \cap \mathbb{Z}^d$ to the symmetric convex body $(K - K) \cap H$ gives a proper symmetric homogeneous rank- r GAP $Q \subseteq H \cap \mathbb{Z}^d$ such that

$$S \subseteq Q, \quad |Q| \ll_d |S|. \quad (*)$$

If $r < d$, then, writing $B_\mathbb{R} = \prod_i [-M_i, M_i]$,

$$|S| \leq |2B_\mathbb{R} \cap \mathbb{Z}^d| = \prod_{i=1}^d (4M_i + 1) \leq 2^d |B|,$$

so (*) gives $|Q| \ll_d |B|$. If $r = d$, then $K - K$, and hence K , is full-dimensional, and the full-rank part of Lemma 5.1, the Rogers–Shephard inequality [6] and (12) give

$$|Q| \ll_d |S| \ll_d \text{vol}_d(K - K) \leq \binom{2d}{d} \text{vol}_d(K) \ll_d \eta \text{vol}_d(B_\mathbb{R}) \ll_d \eta |B|.$$

Finally write $Q = \{\sum_{i=1}^r n_i q_i : |n_i| \leq N_i\}$ and put

$$B' = \prod_{i=1}^r [-N_i, N_i] \cap \mathbb{Z}^r, \quad \phi = \phi_Q|_W, \quad \lambda'(n_1, \dots, n_r) = \sum_{i=1}^r n_i \lambda(q_i),$$

where ϕ_Q is the coefficient identification of the proper GAP Q ; then $|B'| = |Q|$ and ϕ is injective. Lemma 4.3 gives $\lambda'(\phi(v)) = \lambda(v)$ for $v \in W$. The preceding bounds for $|Q|$ give alternatives (i) and (ii), and the identity of functional values proves the final assertion. $\square$

8. THE DENSITY-INCREMENT ITERATION

We use the exponent sequence defined in Section 3.

Lemma 8.1 (Uniform up-move numerology). *Fix $D \geq 2$ and $\zeta_0 > 0$. There are $\varepsilon_0, c_* > 0$ such that, whenever*

$$1 \leq d < e \leq D, \quad \zeta_0 \leq \zeta \leq 1 - \sigma_d, \quad 0 < \varepsilon < \varepsilon_0,$$

one has

$$(\sigma_e + \zeta) \left(\frac{1}{\sigma_d + \zeta} - (1 - \varepsilon)(e - d) \right) \leq 1 - c_*$$

whenever the parenthetical expression is nonnegative.

Proof. For $k = e - d$, set

$$f_{d,k}(\zeta) = (\sigma_{d+k} + \zeta) \left(\frac{1}{\sigma_d + \zeta} - k \right).$$

At $\zeta = 0$, direct substitution gives $f_{d,k}(0) \leq 1$; equality occurs only for

$$(d, e) \in \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

Moreover

$$f'_{d,k}(\zeta) = \frac{\sigma_d - \sigma_{d+k}}{(\sigma_d + \zeta)^2} - k < 0.$$

Hence the required gap is strictly positive at $\varepsilon = 0$ for every pair and every $\zeta \in [\zeta_0, 1 - \sigma_d]$ for which the parenthetical factor is nonnegative. There are finitely many pairs $1 \leq d < e \leq D$; compactness and continuity give a uniform gap $c_* > 0$, which persists after choosing $\varepsilon_0 > 0$ sufficiently small. $\square$

Theorem 8.2 (Projective theorem). *For every fixed d and $\beta > 1$, every (λ, X) -shell non-dividing set $V \subseteq B \subseteq \mathbb{Z}^d$, where B is a symmetric coordinate box and $|B| \leq |V|^\beta$, satisfies*

$$|V| \leq |B|^{\sigma_d + o(1)}.$$

Here the $o(1)$ is uniform over the shell functional and shell parameter once d and β are fixed.

Proof. Assume for a contradiction that for some fixed $\zeta_0 > 0$ there are arbitrarily large examples

$$A_0 \subseteq B_0 \subseteq \mathbb{Z}^{d_0}, \quad |A_0| > |B_0|^{\sigma_{d_0} + \zeta_0}.$$

Write $n_0 = |A_0|$. Since $|A_0| \leq |B_0|$, the assumed examples are already impossible when $\sigma_{d_0} + \zeta_0 \geq 1$; hence we may and do assume $\sigma_{d_0} + \zeta_0 < 1$. All sets constructed below remain shell non-dividing, because the only operations are passage to subsets and coefficient identifications of proper homogeneous GAPs, with the shell functional transported by Lemma 4.3.

Choice of parameters. At any noncontradictory stage $A \subseteq B \subseteq \mathbb{Z}^d$ with excess at least ζ_0 , the inequalities $|A| \leq |B|$ and $|A| > |B|^{\sigma_d + \zeta_0}$ give

$$|B| < |A|^{\beta_*}, \quad \beta_* = (\sigma_1 + \zeta_0)^{-1} > 1.$$

Thus one fixed polynomial-box exponent works at every stage. Since $\sigma_r \rightarrow 1$, the integer

$$D_0 = \max\{r \geq 1 : \sigma_r \leq 1 - \zeta_0\}$$

is finite. At the beginning of every noncontradictory stage, the exact excess is at least ζ_0 and $|A| \leq |B|$, so its ambient dimension is at most D_0 .

For each $1 \leq \ell \leq D_0$, let G_ℓ be a bound for the output dimension in Lemma 4.5 with ambient dimension ℓ and polynomial-box exponent β_* . Choose

$$D > \max\{D_0, G_1, \dots, G_{D_0}\}.$$

Thus every one-step preprocessing output has dimension at most D , while every stage that survives to the next iteration has dimension at most D_0 . Apply Lemma 8.1 with D and ζ_0 , obtaining a uniform up-move gap $c_* > 0$.

Put

$$\Delta_D = \min_{2 \leq r \leq D} (\sigma_r - \sigma_{r-1}) > 0.$$

Choose $\varepsilon > 0$ so small that

$$\varepsilon < \min \left\{ \varepsilon_0, \frac{\zeta_0}{4}, \frac{c_*}{4}, \frac{\Delta_D}{8}, \frac{1}{3} \right\}$$

and so that $\kappa_{r-1} + \varepsilon < \sigma_r + \zeta_0$ for every $2 \leq r \leq D$. Choose K_0 sufficiently large that

$$\frac{1}{K_0} < \sigma_1 + \zeta_0, \quad K_0(\sigma_1 + \zeta_0) > 2.$$

For $1 \leq \ell \leq D_0$, let $K_\ell = K(\ell, \beta_*, \varepsilon, K_0)$ be an admissible exponent in Lemma 4.5, and put

$$K = \max_{1 \leq \ell \leq D_0} K_\ell.$$

Using this larger exponent is valid in every ambient dimension at most D_0 . By the final assertion of Lemma 4.5, there is a fixed number $\beta_{\text{pkg}} > 1$, depending only on the constants chosen so far, which bounds the polynomial-box exponent of every structural package output by preprocessing in dimensions at most D_0 . Put

$$\beta^\dagger = \max \left\{ \beta_{\text{pkg}}, \frac{2\beta_*}{1-\varepsilon} \right\}.$$

Finally choose $0 < \kappa < 1$ sufficiently small in the fixed parameters.

We now choose a slowly varying scale. For each $1 \leq r \leq D-1$, let $N_r(t)$ be a threshold for Lemma 7.1 in dimension r at parameter $\mu = t$. A diagonal choice gives a nondecreasing integer-valued function $\omega(n) \rightarrow \infty$ such that

$$\omega(n)^{2K+2} = o(\log n), \quad N_r(\omega(n)^{-\kappa}) \leq n^{1-\varepsilon} \quad (1 \leq r \leq D-1).$$

For completeness, choose integers $L_m \uparrow \infty$ so that $\log L_m \geq m^{4K+4}$ and $L_m^{1-\varepsilon} \geq \max_r N_r(m^{-\kappa})$, and put $\omega(n) = \max\{m : L_m \leq n\}$ (with any fixed initial value before L_2). No quantitative dependence of $N_r(t)$ on t is required.

At a stage with $n = |A|$, use

$$\delta = \omega(n)^{-1}, \quad \gamma = \delta^K, \quad \mu = \delta^\kappa.$$

Uniformly as $n \rightarrow \infty$, these parameters satisfy

$$\begin{aligned} \delta &\geq n^{-1/2}, & \gamma &\geq n^{-1/3}, & \delta &= o(\mu), \\ \frac{1}{\log n} &= o(\mu\gamma), & \gamma^{-1} &= o(\mu\gamma n), & [s(\delta n)]^{-1} &= o(\gamma). \end{aligned}$$

Each condition follows from $\omega(n) \rightarrow \infty$ together with the growth condition $\omega(n)^{2K+2} = o(\log n)$, which also forces $\omega(n) = n^{o(1)}$; we omit the routine verification, which is carried out in full in the formalization. The parameters therefore meet all hypotheses of the preprocessing theorem, and the last condition in the definition of ω will supply the largeness hypothesis in the weighted convex-density lemma after preprocessing.

The one-step increment. Suppose $A \subseteq B \subseteq \mathbb{Z}^d$ is a current stage, put $n = |A|$, and define its exact excess by

$$p = \frac{\log n}{\log |B|} = \sigma_d + \zeta.$$

The case $|B| = 1$ is already impossible for large n . Since $n \leq |B|$, one has

$$\zeta_0 \leq \zeta \leq 1 - \sigma_d.$$

In particular $\sigma_d \leq 1 - \zeta_0$, so $d \leq D_0$. Apply Lemma 4.5 with the ambient-dimension parameter d , and denote its output by $\tilde{A} \subseteq \tilde{B} \subseteq \mathbb{Z}^e$. Put $\tilde{n} = |\tilde{A}|$. Since $\tilde{n} > n^{1-\varepsilon}$ and $\log \tilde{n} \asymp \log n$, the same parameters satisfy, with $\tilde{n}$ in place of n ,

$$\delta \geq \tilde{n}^{-1/2}, \quad \frac{1}{\log \tilde{n}} = o(\mu\gamma), \quad \gamma^{-1} = o(\mu\gamma\tilde{n}), \quad [s(\delta\tilde{n})]^{-1} = o(\gamma).$$

In every preprocessing alternative one has $|\tilde{B}| \ll n^{\beta_*}$; since $\tilde{n} > n^{1-\varepsilon}$, it follows, after absorbing the fixed implicit constant, that

$$|\tilde{B}| \leq \tilde{n}^{\beta^\dagger}$$

for all sufficiently large n . By the definition of $\beta^\dagger$, the retained structural package arises with polynomial-box exponent at most $\beta^\dagger$. Thus the same package is used in every subsequent application of projective convexity, slab exclusion, filling and the index lemma. Finally,

$$\tilde{n} > n^{1-\varepsilon} \geq N_r(\mu) \quad (1 \leq r \leq D-1),$$

by the defining property of ω . Thus every subsequent use of projective convexity and weighted convex density is legitimate for the post-preprocessing set.

If $e > d$, put

$$a = \frac{1}{p} - (1 - \varepsilon)(e - d).$$

The box estimate gives $|\tilde{B}| \ll n^a$. If $a \leq 0$, this is impossible for large n . Otherwise Lemma 8.1 gives

$$(\sigma_e + \zeta)a \leq 1 - c_*.$$

Since $a \leq \beta_*$, after decreasing a constant $c_{\text{up}} = c_{\text{up}}(D, \zeta_0) > 0$ and absorbing the fixed implicit constant, one obtains

$$|\tilde{A}| > n^{1-\varepsilon} > |\tilde{B}|^{\sigma_e + \zeta + c_{\text{up}}}.$$

Thus an upward dimension change increases the excess by at least the fixed amount c_{up} . Any noncontradictory output satisfies $\sigma_e + \zeta + c_{\text{up}} < 1$, and hence $e \leq D_0$; the stage-dimension invariant is therefore preserved.

If $e < d$, then

$$|\tilde{A}| > n^{1-\varepsilon}, \quad |\tilde{B}| \ll |B|.$$

Since $p \leq 1$ and $\sigma_d - \sigma_e \geq \Delta_D$, the choices above give, after absorbing the fixed constant,

$$|\tilde{A}| > |\tilde{B}|^{\sigma_e + \zeta + c_{\text{down}}}$$

for a fixed $c_{\text{down}} > 0$ depending only on D and ζ_0 .

It remains to consider $e = d$. Write

$$|\tilde{A}| = \rho n, \quad |\tilde{B}| \ll \rho^{K_0} |B|.$$

The pair $(\tilde{A}, \tilde{B})$ is irreducible. Let $\tilde{\lambda}$ and X be the transported shell functional and shell parameter. If $d = 1$, Theorem 6.4 says that its projective indexed family is in μ -convex position. It lies in the zero-dimensional affine space $H_{\tilde{\lambda}}$, whose only nonempty half-space is the whole space; that half-space contains every index. Since $\mu < 1$, this is impossible.

Assume $d \geq 2$. By Lemma 5.8, the family $\pi(\tilde{A})$ affinely spans $H_{\tilde{\lambda}}$ and lies in the convex body

$$\Omega = \{y \in H_{\tilde{\lambda}} : Xy \in \tilde{B}_{\mathbb{R}}\},$$

which has nonempty relative interior. Theorem 6.4 and Lemma 7.1, applied in the Euclidean affine space $H_{\tilde{\lambda}} \cong \mathbb{R}^{d-1}$, followed by Lemma 7.2, give a shell non-dividing set A' with

$$|A'| \geq \rho \eta^{a_d} n, \quad a_d = \kappa_{d-1} + \varepsilon, \quad \eta \in [\mu, \mu^{\hat{\tau}_d}],$$

where

$$\hat{\tau}_d := \tau_{d-1} > 0$$

depends only on d and ε . If cap lifting lowers the dimension to some $f < d$, then its box B' satisfies $|B'| \ll |\tilde{B}| \ll |B|$. Moreover $\eta^{a_d} = n^{-o(1)}$, and hence for sufficiently large n ,

$$|A'| \geq n^{1-2\varepsilon}.$$

The same calculation as in the preceding down case gives

$$|A'| > |B'|^{\sigma_f + \zeta + c_{\text{down}}}.$$

Thus every dimension-lowering cap also produces a fixed excess increment.

Suppose instead that cap lifting preserves dimension d . Then

$$|B'| \ll \eta \rho^{K_0} |B|.$$

At the old exponent $p = \sigma_d + \zeta$ we have

$$\frac{|A'|}{|B'|^p} \gg \eta^{a_d - p} \rho^{1 - K_0 p}.$$

The choices of ε and K_0 give, uniformly for $2 \leq d \leq D$,

$$p - a_d \geq \nu_0 > 0, \quad K_0 p - 1 \geq 1.$$

Let

$$\tau_* = \min_{2 \leq r \leq D} \hat{\tau}_r > 0.$$

Since $\eta \leq \mu^{\hat{\tau}_d}$ and $\rho \leq 1$, the right-hand side is at least

$$C_D^{-1} \mu^{-\tau_* \nu_0},$$

which tends to infinity. Therefore, for all sufficiently large n ,

$$|A'| > |B'|^p.$$

The exact excess strictly increases in a same-dimensional step. In addition,

$$\frac{|B'|}{|B|} \ll_D \eta \rho^{K_0} \leq C_D \mu^{\tau_*} = o(1).$$

This completes the one-step analysis.

The iteration terminates. We first prove that the iteration cannot lose too many points. An upward step increases the excess by at least c_{up} , while the exact excess is always at most $1 - \sigma_d < 1$. Hence there are only $O_{D, \zeta_0}(1)$ upward steps. Every downward step decreases the integer dimension, and between two upward steps there can be at most D downward steps. Since all dimensions are at most D , the total number of dimension-changing steps is at most some fixed number $M = M(D, \zeta_0)$.

For $2 \leq r \leq D$ put

$$c_r = \max \left\{ \kappa_{r-1} + \varepsilon, \frac{1}{K_0} \right\}.$$

By construction

$$c_r < \sigma_r + \zeta_0.$$

Consider any consecutive block of same-dimensional steps in dimension r . At a step in this block, let ρ_j be the preprocessing cardinality ratio and let η_j be the cap-volume ratio. Put

$$x_j = \log(1/\eta_j), \quad y_j = \log(1/\rho_j).$$

The point and box estimates give

$$\begin{aligned} \log \frac{|A_j|}{|A_{j+1}|} &\leq (\kappa_{r-1} + \varepsilon)x_j + y_j, \\ \log \frac{|B_j|}{|B_{j+1}|} &\geq x_j + K_0 y_j - O_D(1). \end{aligned}$$

The first right-hand side is at most $c_r(x_j + K_0 y_j)$. Also

$$x_j \geq \tau_* \log(1/\mu_j) = \tau_* \kappa \log \omega(|A_j|) \longrightarrow \infty$$

uniformly whenever all current cardinalities are bounded below by a fixed positive power of n_0 , because ω is nondecreasing and tends to infinity. Thus the $O_D(1)$ term is $o(x_j + K_0 y_j)$ uniformly. Summing over any initial segment of the block gives

$$\log \frac{|A_{\text{start}}|}{|A_{\text{end}}|} \leq (c_r + o(1)) \log |B_{\text{start}}|.$$

Since the starting excess is at least ζ_0 ,

$$|A_{\text{end}}| \geq |A_{\text{start}}|^{1-c_r/(\sigma_r+\zeta_0)-o(1)}.$$

Therefore there is a fixed $\theta > 0$, depending only on the chosen parameters, such that every same-dimensional block, including every partial block, retains at least the θ -th power of its input cardinality. Every dimension-changing step retains at least $|A_j|^{1-2\varepsilon}$ points.

There are at most M dimension-changing steps and at most $M+1$ same-dimensional blocks. Hence there is a fixed $q > 0$ such that every iterate produced before a contradiction satisfies

$$|A_j| \geq n_0^q.$$

To remove the apparent circularity in the uniform $o(1)$ above, run the construction with the provisional stopping rule $|A_j| < n_0^{q/2}$. On the range before this rule triggers, all $o(1)$ estimates are uniform; the preceding product calculation yields the stronger lower bound $|A_j| \geq n_0^q$, so the provisional rule never triggers.

We may therefore iterate indefinitely unless a contradiction occurs. There are only finitely many dimension changes. After the last one, every step has the same dimension, and the box estimate above gives, uniformly for all remaining stages and sufficiently large n_0 ,

$$|B_{j+1}| \leq \frac{1}{2}|B_j|.$$

A positive integer box cardinality cannot be halved indefinitely, while the accompanying sets all have cardinality at least $n_0^q > 1$. This is the required contradiction. $\square$

9. PROOF OF THE MAIN THEOREM

Proof of Theorem 1.2. The lower bound follows from Proposition 2.2. For the upper bound, fix $0 < \varepsilon < 1$ and suppose, for arbitrarily large N , that there is a non-dividing set $A \subseteq [N]$ with

$$|A| > N^{1/5+2\varepsilon}.$$

Partition $[N]$ into dyadic intervals. Some interval $[X, 2X)$ contains a subset $A_0 \subseteq A$ with

$$|A_0| \geq \frac{|A|}{1 + \log_2 N} > N^{1/5+\varepsilon}$$

for all sufficiently large N . Take

$$d = 1, \quad B = [-2X, 2X] \cap \mathbb{Z}, \quad \lambda(t) = t.$$

Then A_0 is shell non-dividing and $|B| \leq 4N + 1$. Consequently, for all sufficiently large N ,

$$|A_0| > |B|^{1/5+\varepsilon/2}, \quad |B| < |A_0|^\beta, \quad \beta = (1/5 + \varepsilon/2)^{-1} > 1.$$

Theorem 8.2, applied with this fixed β , gives

$$|A_0| \leq |B|^{1/5+o(1)}.$$

For sufficiently large N the $o(1)$ term is smaller than $\varepsilon/2$, contradicting the preceding strict inequality. Hence $F(N) \leq N^{1/5+o(1)}$, and the theorem follows. $\square$

10. A CONCLUDING REMARK

For $x = \min A$, a residue fiber $A_r = \{a \in A : a \equiv r \pmod{x}\}$ of a non-dividing set $A \subseteq [N]$ becomes a non-averaging set after the affine normalization $a \mapsto (a - r)/x$. If a fiber has size h , the sharp non-averaging bound therefore gives $N \geq xh^{4-o(1)}$, so a direct multiplicity amplification estimate of the form

$$h \geq |A|^{5/4-o(1)}x^{-1/4}$$

would give another proof of Theorem 1.2. Such a statement is false for arbitrary zero-sum-free sequences, so it would have to exploit the simultaneous forbidden-modulus property.

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